KI-HOON LEE

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Portfolio: ki-hoon-lee.github.io

EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY | Berkeley, CA | MEng Mechanical Engineering | 3.83/4.00 Aug 2024 - May 2025
UNIVERSITY OF BATH | Bath, UK | BEng in Mechanical Engineering | 3.70/4.00 equivalent Sep 2019 - May 2024

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer | ZoNexus LLC | Richmond, CA Jul 2025 – Present

- Owned full product lifecycle for precision mechanical subsystems from concept through production, delivering functional prototypes in a fast-paced startup environment through iterative design, fabrication, and testing.
- Collaborated with Lawrence Berkeley National Lab researchers to rapidly iterate hardware designs based on experimental feedback, reducing development cycle time and improving system performance.
- Integrated electromechanical assemblies including sensor arrays, stepper motor systems, and custom PCBs into unified vacuum-compatible platforms achieving angstrom level positioning precision.
- Designed components for manufacturability using GD&T-compliant drawings, partnering with machinists and suppliers to achieve tight-tolerance fabrication (<0.0002" where critical) and repeatable assembly.
- Established material specifications, thermal management strategies, and electrical routing standards to ensure reliable operation under vacuum, cryogenic, and high-thermal-load conditions.
- Created assembly documentation and validation protocols that standardized prototype builds and enabled transition to production-scale manufacturing.

Product Design Engineer (Capstone) | Blue Goji | Berkeley, CA Sep 2024 – May 2025

- Designed and prototyped a resistive omnidirectional treadmill for medical rehabilitation and VR applications, integrating complex electromechanical drive systems for multi-directional motion.
- Reduced Bill of Materials (BOM) by 50% through in-house fabrication of custom components via CNC machining and 3D printing, optimizing part geometry for manufacturability.
- Validated structural integrity and clinical safety through iterative Finite Element Analysis (FEA) and human-subject testing, achieving target Factor of Safety (FoS) margins.

PROJECTS

Peltier Cooler with PID Temp Controller | ZoNexus LLC 2026

- Designed an active thermal regulation system capable of reaching below -60 °C for cryo-testing by integrating a multi-layer Peltier stack with water-cooled heat exchangers capable of operating in vacuum.
- Developed a dedicated PID controller to maintain thermal stability within ±0.1°C under varying external loads.

Air Free Transfer Motorized Holder & Stepper Controller | ZoNexus LLC 2025

- Engineered a multi-sample motorized holder featuring a hermetically sealed tip chamber to maintain sample integrity during transfer between ambient, argon filled glove box, and ultra-high vacuum (UHV) environments.
- Developed a custom stepper motor controller and PCB from the ground up, achieving sub 60-angstrom positioning precision verified through a Transmission Electron Microscope (TEM) imaging.

GoalE: Interactive Ball Tracking Embedded System | UC Berkeley: ME235 | Embedded Systems Project 2025

- Developed a real-time multitasking system using an ESP32, ultrasonic sensors, and servo actuators to achieve autonomous position tracking and interception with live LabVIEW telemetry.

Soccer Player Market Value Predictor | UC Berkeley: ME249 | ML Project 2024

- Built a Keras neural network to predict EPL soccer player market values from 5,000+ player FIFA24 dataset; applied PCA to reduce 41 features to 7 components retaining >90% variance, achieving MAE of $784K.

SKILLS

Mechanical & Design: SolidWorks, Autodesk (Fusion, AutoCAD, Nastran), GD&T, DFM/DFA/DOE, CNC Machining, 3D Printing
Electrical & Hardware: PCB Design (KiCAD, EasyEDA), Embedded Systems, Soldering, Sensors & Actuators, Thermal Analysis
Software & Control: Python (TensorFlow/Keras, Pandas), ML, MATLAB/Simulink, LabVIEW, PID Control, C++ (Arduino/ESP32)